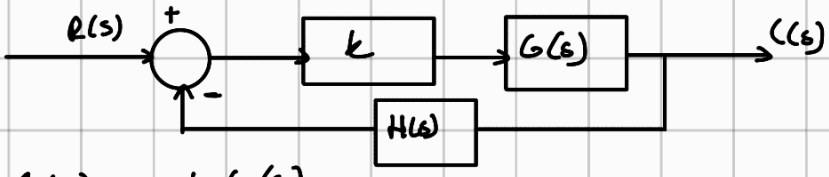
Root locus



$$\frac{C(s)}{R(s)} = \frac{kG(s)}{1+kG(s)H(s)}$$

$1 + kG(s)H(s) = 0 \Rightarrow G(s)H(s) = -\frac{1}{k}$

- mag: $|G(s)H(s)| = \frac{1}{k}$
- phase: $\angle G(s)H(s) = \pm 180(2i+1)$

$\begin{cases} k = 0 \Rightarrow \text{poles of the OLTF} \\ k \uparrow \infty \Rightarrow \text{zero} \;\; // \;\; // \;\; // \end{cases}$

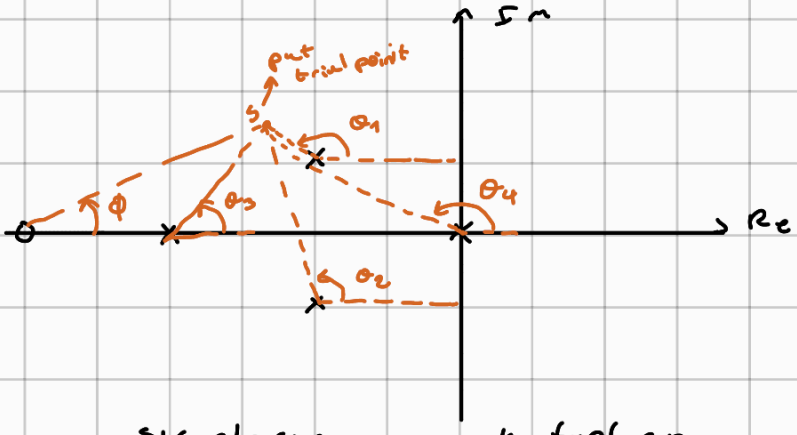
n: number of poles
m: // // zeros

$\begin{cases} n: \text{branches} \\ n-m: \text{asymptotes} \end{cases}$

$$\phi = \frac{\pm 180(2i+1)}{n-m}$$

Centroid: $\frac{\Sigma \text{Real poles} - \Sigma \text{Real zeros}}{n-m}$

s - Complex conjugate poles



sıfırların açılar toplamı − kutupların açılar toplamı $= \pm 180^\circ$

Step 6 Intersection with jw axis